

Technology Stack

Date	29.09.2026
Team ID	UNM123APEXLAB
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript (Jinja2 templates)	Lightweight and responsive UI for entering story prompts and viewing generated comic panels on any device. No heavy framework is needed, so pages load fast and are easy to maintain.
2	Backend / Server-Side	Python with FastAPI (Uvicorn server)	FastAPI gives high-performance async APIs, automatic validation and built-in docs. Python is the natural fit for calling Gemini AI models and handling image and text processing.
3	Database / Data Storage	SQLite with SQLAlchemy ORM	Simple, serverless and zero-config storage for user prompts, generated stories and image references. SQLAlchemy keeps the code clean and makes a later move to PostgreSQL easy.
4	Cloud / Hosting / Deployment	Docker + Render	Docker packages the app so it runs the same everywhere. Render offers easy deployment from GitHub, HTTPS by default and a free tier, which keeps hosting cost-effective.
5	Version Control & CI/CD	Git, GitHub and GitHub Actions	GitHub enables seamless team collaboration and code review. GitHub Actions runs automated tests and auto-deploys on every push to the main branch.

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
6	Third-Party APIs / Other Tools	Google Gemini API (Gemini text and image models), Google AI Studio	Gemini text models generate the comic storyline, dialogue and panel descriptions from the user's prompt. Gemini image models create the matching illustrations for each panel.